Sparse Precoder Design for Massive MIMO LEO Satellite Communications

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***Abstract*—In this paper, we investigate precoder design for massive multiple-input multiple-output (MIMO) low-earth-orbit (LEO) satellite communication (SATCOM). We first establish a beam based channel model that characterizes the spatial single path property of the LEO SATCOM channel, and reveal its sparsity in the beam domain. Based on this channel model, we prove that the design of the space domain precoder can be simplified into that of a lower-dimensional beam domain precoder. More importantly, we introduce a sparsity constraint on the beam domain precoder and propose a novel sparse precoder, achieving a flexible trade-off between transmission performance and computational complexity. Simulation results demonstrate the satisfactory performance of the proposed sparse precoder with low complexity.**

***Index Terms*—LEO satellite communication, massive MIMO, precoder design, sparse precoder.**

I. Introduction

Satellite communication (SATCOM) has emerged as a pivotal technology to enable global connectivity, particularly in remote and underserved regions where terrestrial networks are limited or unavailable [1], [2]. Low-earth-orbit (LEO) satellites, operated at altitudes between 200 km and 2000 km, have attracted intensive research interest due to reduced latency, lower path loss, and decreased deployment costs compared to the geostationary-earth-orbit (GEO) satellites [3]. These capabilities make LEO satellites a critical component in the evolution toward future integrated space–terrestrial networks, supporting seamless coverage and meeting the growing demands for high-throughput, reliable communication worldwide. Meanwhile, a key technology for the fifth-generation cellular wireless communications and its future evolution, massive multiple-input multiple-output (MIMO), has been introduced into SATCOM to enhance connectivity, reliability, and data rates by deploying a large number of antennas at the satellite and serving a number of user terminals (UTs) in the same time-frequency resource simultaneously [4].

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Extensive research efforts have been devoted to the design of uplink (UL) and downlink (DL) transmissions for massive MIMO LEO SATCOM systems [4]–[9]. For example, due to the long propagation delay and mobility of satellites and UTs, the instantaneous channel state information (iCSI) at the transmitter sides of SATCOM systems is typically hard to acquire. Thus the statistical CSI (sCSI) based DL precoder is proposed in [4] to maximize the average signal-to-leakage-plus-noise ratio (ASLNR). In [5], the single-stream precoding for each UT is proved to be optimal for maximizing the ergodic sum-rate, even when UTs are equipped with mutiple antennas. Robust transmission is investigated for massive MIMO LEO SATCOM in [6] and [7], considering the security and quality of service (QoS), respectively. Moreover, robust precoding is also investigated in integrated communication and localization (ICAL) systems [8] and cooperative satellite-terrestrial networks [9]. However, in most existing works, the precoders are designed in the space domain with exorbitant computational complexity due to the large-scale array and are not conducive to on-board payload implementation in massive MIMO LEO SATCOM systems.

In this paper, we investigate the DL precoding for massive MIMO LEO SATCOM. We first establish a beam based channel model to characterize the spatial single path property of LEO SATCOM channel and reveal its sparsity in the beam domain. Leveraging the channel model, we prove that the space domain precoder design can be transformed into that of a lower-dimensional beam domain precoder design. Moreover, a novel sparse precoder design is developed by introducing a sparsity constraint on the beam domain precoder, thus achieving a trade-off between performance and complexity.

II. System Model

A. System Configuration

Consider a massive MIMO LEO satellite communication system, where a satellite serves K single-antenna ground UTs simultaneously. The satellite is equipped with a uniform planar array (UPA) with $M = M_{\mathrm{x}}M_{\mathrm{y}}$ antennas incorporating M_{x} antennas along x-axis and M_{y} antennas along y-axis, and the

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antenna spacing is set to half wavelength. The index set of all served UTs is defined as $\mathcal{K} = \{k \,|k = 1, 2, \cdots, K\}$.

The orthogonal frequency division multiplexing (OFDM) modulation is adopted to support the wideband transmission. The number of subcarriers and the length of the cyclic prefix (CP) are denoted by N_{c} and N_{p}, respectively. The sampling interval length is T_{s} and the duration of each OFDM symbol with CP is $T = (N_{\mathrm{c}} + N_{\mathrm{p}})\, T_{\mathrm{s}}$.

B. Beam Based Channel Model

Since the angular spread at the satellite side tends to zero, the LEO channel has the spatial single path property. Specifically, after appropriate frequency and time synchronizations at the UT side, the DL channel between the satellite and UT k at a specific subcarrier is given by [10]

$$\mathbf{h}_k^{\mathrm{H}} = \alpha_k \mathbf{g}_k^{\mathrm{H}}, \tag{1}$$

where $\alpha_k = \sum_{\ell=0}^{L_k-1} \hat{\alpha}_{k,\ell}$, L_k denotes the number of paths and $\hat{\alpha}_{k,\ell}$ is the effective channel gain associated with the path ℓ of the channel between the satellite and UT k. $\mathbf{g}_k$ is the steering vector at the satellite for UT k, which is defined as

$$\mathbf{g}_k = \mathbf{v}_{\mathrm{x}}\left(g_k^{\mathrm{x}}\right) \otimes \mathbf{v}_{\mathrm{y}}\left(g_k^{\mathrm{y}}\right), \tag{2}$$

where $\otimes$ denotes the Kronecker product operator, $g_k^{\mathrm{x}} = \sin\varphi_k^{\mathrm{y}} \cos\varphi_k^{\mathrm{x}}$ and $g_k^{\mathrm{y}} = \cos\varphi_k^{\mathrm{y}}$ are the directional cosines of UT k with respect to (w.r.t.) x-axis and y-axis, and φ_k^{x} and φ_k^{y} are the corresponding angles w.r.t. x-axis and y-axis, respectively. Let $\mathrm{t} \in \{\mathrm{x}, \mathrm{y}\}$, then $\mathbf{v}_{\mathrm{t}}(g)$ in (2) is defined as

$$\mathbf{v}_{\mathrm{t}}(g) = \frac{1}{\sqrt{M_{\mathrm{t}}}}\left[1, e^{-\bar{\jmath}\pi g}, \cdots, e^{-\bar{\jmath}\pi(M_{\mathrm{t}}-1)g}\right]^{\mathrm{T}} \in \mathbb{C}^{M_{\mathrm{t}}}. \tag{3}$$

Note that the angle w.r.t. t-axis, φ_k^{t} satisfies $\varphi_k^{\mathrm{t}} \in [\pi/2, \pi/2]$. Then the corresponding directional cosine g_k^{t} satisfies $g_k^{\mathrm{t}} \in [-1, 1]$. Uniformly divide this set into multiple disjoint subsets as $\Lambda_{n_{\mathrm{t}}} = \left[\tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}}, \tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}} + 2/(F_t M_{\mathrm{t}})\right)$, where $\tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}} = 2n_{\mathrm{t}}/(F_t M_{\mathrm{t}}) - \tilde{g}_{\max}$, $n_{\mathrm{t}} \in \{0, 1, \cdots, N_{\mathrm{t}} - 1\}$, $N_{\mathrm{t}} = \lceil \tilde{g}_{\max} F_{\mathrm{t}} M_{\mathrm{t}} \rceil$ is the sampling number, and F_{t} is the fine factor. We use $\tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}}$ to approximate the directional cosines in set $\Lambda_{n_{\mathrm{t}}}$ and define the beam matrix of t-axis as

$$\mathbf{V}_{\mathrm{t}} = \left[\tilde{\mathbf{v}}_{\mathrm{t}}\left(\tilde{g}_0^{\mathrm{t}}\right), \tilde{\mathbf{v}}_{\mathrm{t}}\left(\tilde{g}_1^{\mathrm{t}}\right), \cdots, \tilde{\mathbf{v}}_{\mathrm{t}}\left(\tilde{g}_{N_{\mathrm{t}}-1}^{\mathrm{t}}\right)\right] \in \mathbb{C}^{M_{\mathrm{t}} \times N_{\mathrm{t}}}, \tag{4}$$

where

$$\tilde{\mathbf{v}}_{\mathrm{t}}\left(\tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}}\right) = \frac{1}{\sqrt{M_{\mathrm{t}}}}\left[1, e^{-\bar{\jmath}\pi \tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}}}, \cdots, e^{-\bar{\jmath}\pi(M_{\mathrm{t}}-1)\tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}}}\right]^{\mathrm{T}} \in \mathbb{C}^{M_{\mathrm{t}}} \tag{5}$$

is the sampling steering vector. Each column of the beam matrix $\mathbf{V}_{\mathrm{t}}$ corresponds to a specific beam pointing towards a sampling directional cosine. Then $\mathbf{v}_{\mathrm{t}}(g)$ in (2) becomes

$$\mathbf{v}_{\mathrm{t}}\left(g_k^{\mathrm{t}}\right) = \mathbf{V}_{\mathrm{t}} \tilde{\mathbf{g}}_k^{\mathrm{t}}, \tag{6}$$

where $\tilde{\mathbf{g}}_k^{\mathrm{t}}$ is the diffusion vector, which represents the energy distribution across multiple adjacent beams. Specifically, when the directional cosine of a path is equal to a specific sampling directional cosine, i.e., $g_k^{\mathrm{t}} = \tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}}$, we have $\tilde{\mathbf{g}}_k^{\mathrm{t}} = \mathbf{e}_{n_{\mathrm{t}}}$, where the n_{t}-th element of $\mathbf{e}_{n_{\mathrm{t}}}$ is 1 and all the other elements are 0, indicating that all the energy is concentrated in the n_{t}-th beam. When g_k^{t} is between two sampling directional cosines, i.e., $\tilde{g}_{n_{\mathrm{t}}}^{\mathrm{t}} < g_k^{\mathrm{t}} < \tilde{g}_{n_{\mathrm{t}}+1}^{\mathrm{t}}$, the energy is typically spread out over multiple beams, and the compressive sensing technique can be used to acquire the diffusion vector $\tilde{\mathbf{g}}_k^{\mathrm{t}}$ according to (6). Then (2) can be rewritten as

$$\mathbf{g}_k = \left(\mathbf{V}_{\mathrm{x}} \otimes \mathbf{V}_{\mathrm{y}}\right)\left(\tilde{\mathbf{g}}_k^{\mathrm{x}} \otimes \tilde{\mathbf{g}}_k^{\mathrm{y}}\right) = \mathbf{V}\tilde{\mathbf{g}}_k, \tag{7}$$

where $\mathbf{V} = \mathbf{V}_{\mathrm{x}} \otimes \mathbf{V}_{\mathrm{y}} \in \mathbb{C}^{M \times N}$, $\tilde{\mathbf{g}}_k = \tilde{\mathbf{g}}_k^{\mathrm{x}} \otimes \tilde{\mathbf{g}}_k^{\mathrm{y}} \in \mathbb{C}^N$, and $N = N_{\mathrm{x}} N_{\mathrm{y}}$. Then the DL channel in (1) can be rewritten as

$$\mathbf{h}_k^{\mathrm{H}} = \tilde{\mathbf{h}}_k^{\mathrm{H}} \mathbf{V}^{\mathrm{H}}, \tag{8}$$

where $\tilde{\mathbf{h}}_k = \alpha_k \tilde{\mathbf{g}}_k$ denotes the beam domain channel and the channel model in (8) is referred to as the beam based channel model. Note that most elements in diffusion vector $\tilde{\mathbf{g}}_k^{\mathrm{t}}$ are approximately zero, which means that the channel exhibits significant sparsity in the beam domain. Thus, we define $\mathcal{N}_k = \left\{n_1 < n_2 < \cdots < n_{N_k} | [\tilde{\mathbf{h}}_k]_{n_i} \neq 0, 1 \leq i \leq N_k\right\}$ as the ordered index set of non-zero beam domain channel elements with $N_k = |\mathcal{N}_k|$. Then the channel in (8) becomes

$$\mathbf{h}_k^{\mathrm{H}} = \left[\tilde{\mathbf{h}}_k\right]_{\mathcal{N}_k}^{\mathrm{H}} \mathbf{S}_{\mathcal{N}_k}^{\mathrm{H}} \mathbf{V}^{\mathrm{H}}, \tag{9}$$

where $\mathbf{S}_{\mathcal{N}_k} \in \mathbb{C}^{N \times N_k}$ denotes the beam selection matrix with

$$\left[\mathbf{S}_{\mathcal{N}_k}\right]_{i,j} = \begin{cases} 1, & i = [\mathcal{N}_k]_j, \\ 0, & \text{otherwise}. \end{cases} \tag{10}$$

Further, we present statistical properties of satellite channels. Assume that the channel gain α_k is Rician distribution with Rician factor κ_k and the average channel power $\mathbb{E}\left\{|\alpha_k|^2\right\} = \beta_k$. Then the channel correlation matrix at the satellite is given by

$$\mathbf{R}_k^{\mathrm{sat}} = \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^{\mathrm{H}}\right\} = \beta_k \mathbf{V}\tilde{\mathbf{g}}_k \tilde{\mathbf{g}}_k^{\mathrm{H}} \mathbf{V}^{\mathrm{H}}. \tag{11}$$

The matrix $\mathbf{R}_k^{\mathrm{sat}}$ is rank-one due to the spatial single path property, which indicates that the satellite channels are highly correlated in the space domain at the satellite side. Assume that the satellite has sCSI, which is utilized for precoder design.

III. Sparse Precoder Design

A. Beam Structured Precoder

The received signal at UT k is given by

$$y_k = \mathbf{h}_k^{\mathrm{H}} \sum_{i \in \mathcal{K}} \mathbf{w}_i x_i + z_k, \tag{12}$$

where $\mathbf{w}_i$ is the DL precoder for UT i, x_i is the transmit date stream for UT i with zero mean and unit variance, and $z_k \sim \mathcal{CN}\left(0, \sigma_k^2\right)$ denotes the additive white Gaussian noise.

The DL ergodic rate of UT k is given by

$$\mathcal{I}_k = \mathbb{E}\left\{\log\left(1 + \frac{\mathbf{h}_k^{\mathrm{H}} \mathbf{w}_k \mathbf{w}_k^{\mathrm{H}} \mathbf{h}_k}{\sum\limits_{i \in \mathcal{K} \setminus k} \mathbf{h}_k^{\mathrm{H}} \mathbf{w}_i \mathbf{w}_i^{\mathrm{H}} \mathbf{h}_k + \sigma_k^2}\right)\right\}. \tag{13}$$

The DL ergodic sum-rate maximization problem can be formulated as

$$\max_{\{\mathbf{w}_k\}_{k \in \mathcal{K}}} \sum_{k \in \mathcal{K}} \mathcal{I}_k, \text{ s.t. } \sum_{k \in \mathcal{K}} \|\mathbf{w}_k\|^2 \leq P. \tag{14}$$

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Owing to the expectation in the ergodic rate $\mathcal{I}_k$, the precoders need to be calculated using Monte-Carlo method with an exhaustive sample average, which is computationally intensive and hinders the solution. To avoid the computationally cumbersome sample average operations, we approximate the ergodic rate with its upper bound. Specifically, the ergodic rate of UT k can be upper bounded as [5]

$$\mathcal{I}_k \leq \log\left(1+\frac{\beta_k \mathbf{g}_k^{\mathrm{H}}\mathbf{w}_k\mathbf{w}_k^{\mathrm{H}}\mathbf{g}_k}{\sum\limits_{i\in\mathcal{K}\setminus k}\beta_k\mathbf{g}_k^{\mathrm{H}}\mathbf{w}_i\mathbf{w}_i^{\mathrm{H}}\mathbf{g}_k+\sigma_k^2}\right)=\mathcal{I}_k^{\mathrm{ub}}. \quad (15)$$

Then the original problem in (14) can be approximated as

$$\max_{\{\mathbf{w}_k\}_{k\in\mathcal{K}}}\sum_{k\in\mathcal{K}}\mathcal{I}_k^{\mathrm{ub}},\ \text{s.t.}\ \sum_{k\in\mathcal{K}}\|\mathbf{w}_k\|^2\leq P. \quad (16)$$

Note that only sCSI is used in the problem (16). The required sCSI includes the average channel power and the steering vectors, i.e., β_k and $\mathbf{g}_k, \forall k$. The following theorem reveals a key insight, whose detailed proof is omitted due to page limitations.

Theorem 1: When $M\to\infty$, the optimal solution of problem in (16) can be obtained as

$$\mathbf{w}_k^{\mathrm{op}}=\mathbf{V}\mathbf{S}_{\mathcal{N}_k}\tilde{\mathbf{w}}_k^{\mathrm{op}},\ \forall k\in\mathcal{K}. \quad (17)$$

In (17), $\tilde{\mathbf{w}}_k^{\mathrm{op}}$ is obtained as

$$\{\tilde{\mathbf{w}}_k^{\mathrm{op}}\}_{k\in\mathcal{K}}=\arg\max_{\{\tilde{\mathbf{w}}_k\}_{k\in\mathcal{K}}}\sum_{k\in\mathcal{K}}\tilde{\mathcal{I}}_k^{\mathrm{ub}},\ \text{s.t.}\ \sum_{k\in\mathcal{K}}\|\tilde{\mathbf{w}}_k\|^2\leq P, \quad (18)$$

where

$$\tilde{\mathcal{I}}_k^{\mathrm{ub}}=\log\left(1+\frac{\beta_k\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{I}_{k,k}\tilde{\mathbf{w}}_k\tilde{\mathbf{w}}_k^{\mathrm{H}}\mathbf{I}_{k,k}\tilde{\mathbf{g}}_k}{\sum\limits_{i\in\mathcal{K}\setminus k}\beta_k\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{I}_{k,i}\tilde{\mathbf{w}}_i\tilde{\mathbf{w}}_i^{\mathrm{H}}\mathbf{I}_{i,k}\tilde{\mathbf{g}}_k+\sigma_k^2}\right) \quad (19)$$

and $\mathbf{I}_{k,i}=\mathbf{S}_{\mathcal{N}_k}^{\mathrm{H}}\mathbf{S}_{\mathcal{N}_i}$.

Theorem 1 reveals that for a sufficiently large M, the design of the optimal space domain precoder $\mathbf{w}_k^{\mathrm{op}}$ can be transformed into that of the beam domain precoder $\tilde{\mathbf{w}}_k^{\mathrm{op}}$. Due to the sparsity of beam domain channels in massive MIMO SATCOM system, the dimension of the beam domain precoder N_k is typically much smaller than that of the space domain precoder M.

Although the number of satellite antennas in practical system is finite, the beam structured precoder in Theorem 1 can still be employed to achieve near-optimal performance while maintaining low complexity, particularly when the satellite is equipped with a large number of antennas. To this end, we restrict the space domain precoder for UT u to satisfy the beam structure as follows

$$\mathbf{w}_k=\mathbf{V}\mathbf{S}_{\mathcal{N}_k}\tilde{\mathbf{w}}_k. \quad (20)$$

Then the ergodic rate upper bound in (15) becomes

$$\mathcal{I}_k^{\mathrm{b,ub}}=\log\left(1+\frac{\beta_k\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{Q}_{k,k}\tilde{\mathbf{w}}_k\tilde{\mathbf{w}}_k^{\mathrm{H}}\mathbf{Q}_{k,k}^{\mathrm{H}}\tilde{\mathbf{g}}_k}{\sum\limits_{i\in\mathcal{K}\setminus k}\beta_k\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{Q}_{k,i}\tilde{\mathbf{w}}_i\tilde{\mathbf{w}}_i^{\mathrm{H}}\mathbf{Q}_{k,i}^{\mathrm{H}}\tilde{\mathbf{g}}_k+\sigma_k^2}\right), \quad (21)$$

where $\mathbf{Q}_{k,i}=\mathbf{S}_{\mathcal{N}_k}^{\mathrm{H}}\mathbf{V}^{\mathrm{H}}\mathbf{V}\mathbf{S}_{\mathcal{N}_i}$. We turn to optimize the beam domain precoder $\tilde{\mathbf{w}}_k$ to maximize the ergodic sum-rate upper bound as follows

$$\max_{\{\tilde{\mathbf{w}}_k\}_{k\in\mathcal{K}}}\sum_{k\in\mathcal{K}}\mathcal{I}_k^{\mathrm{b,ub}},\ \text{s.t.}\ \sum_{k\in\mathcal{K}}\tilde{\mathbf{w}}_k^{\mathrm{H}}\mathbf{A}_k\tilde{\mathbf{w}}_k\leq P, \quad (22)$$

where $\mathbf{A}_k=\mathbf{S}_{\mathcal{N}_k}^{\mathrm{H}}\mathbf{V}^{\mathrm{H}}\mathbf{V}\mathbf{S}_{\mathcal{N}_k}$. The problem in (22) can be transformed into the following weighted minimum mean-squared error (WMMSE) minimization problem [11]:

$$\begin{aligned}&\min_{\{\tilde{\mathbf{w}}_k,\tilde{q}_k,\tilde{\gamma}_k\}_{k\in\mathcal{K}}}\sum_{k\in\mathcal{K}}\tilde{q}_k\tilde{\varepsilon}_k-\log\left(\tilde{q}_k\right)\\&\text{s.t.}\sum_{k\in\mathcal{K}}\tilde{\mathbf{w}}_k^{\mathrm{H}}\mathbf{A}_k\tilde{\mathbf{w}}_k\leq P,\end{aligned} \quad (23)$$

where

$$\begin{aligned}\tilde{\varepsilon}_k=&1-\tilde{\gamma}_k\sqrt{\beta_k}\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{Q}_{k,k}\tilde{\mathbf{w}}_k-\sqrt{\beta_k}\tilde{\mathbf{w}}_k^{\mathrm{H}}\mathbf{Q}_{k,k}^{\mathrm{H}}\tilde{\mathbf{g}}_k\tilde{\gamma}_k^*\\&+|\gamma_k|^2\left(\beta_k\sum_{i\in\mathcal{K}}\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{Q}_{k,i}\tilde{\mathbf{w}}_i\tilde{\mathbf{w}}_i^{\mathrm{H}}\mathbf{Q}_{k,i}^{\mathrm{H}}\tilde{\mathbf{g}}_k+\sigma_k^2\right).\end{aligned} \quad (24)$$

Then the beam structured precoder can be obtained using the WMMSE in Algorithm 1.

Algorithm 1: Beam Structured Precoder

Input: $\beta_k, \tilde{\mathbf{g}}_k, \mathbf{A}_k, \sigma_k^2, \mathbf{Q}_{k,k'}, \forall k,k'\in\mathcal{K}, P$

Initialize: $\tilde{\mathbf{w}}_k$.

repeat

$\forall k$, update $\tilde{\gamma}_k$:

$$\tilde{\gamma}_k=\left(\beta_k\sum_{i\in\mathcal{K}}\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{Q}_{k,i}\tilde{\mathbf{w}}_i\tilde{\mathbf{w}}_i^{\mathrm{H}}\mathbf{Q}_{k,i}^{\mathrm{H}}\tilde{\mathbf{g}}_k+\sigma_k^2\right)^{-1}\times\sqrt{\beta_k}\tilde{\mathbf{w}}_k^{\mathrm{H}}\mathbf{Q}_{k,k}^{\mathrm{H}}\tilde{\mathbf{g}}_k$$

$\forall k$, update $\tilde{q}_k$:

$$\tilde{q}_k=\left(1-\tilde{\gamma}_k\sqrt{\beta_k}\tilde{\mathbf{g}}_k^{\mathrm{H}}\mathbf{Q}_{k,k}\tilde{\mathbf{w}}_k\right)^{-1}$$

$\forall k$, update $\tilde{\mathbf{w}}_k$:

$$\tilde{\mathbf{w}}_k=\left(\sum_{i\in\mathcal{K}}\tilde{q}_i|\gamma_i|^2\beta_i\mathbf{Q}_{i,k}^{\mathrm{H}}\tilde{\mathbf{g}}_i\tilde{\mathbf{g}}_i^{\mathrm{H}}\mathbf{Q}_{i,k}+\mu\mathbf{A}_k\right)^{-1}\times\tilde{q}_k\tilde{\gamma}_k\sqrt{\beta_k}\mathbf{Q}_{k,k}^{\mathrm{H}}\tilde{\mathbf{g}}_k$$

until *convergence*;

Output: The space domain precoder is obtained as $\mathbf{w}_k=\mathbf{V}\mathbf{S}_{\mathcal{N}_k}\tilde{\mathbf{w}}_k, \forall k\in\mathcal{K}$.

B. Sparse Precoder Design

In the beam structured precoder designs, the beams are heuristically predetermined for each UT, which inevitably restricts the design space and degrades the system performance. To overcome this limitation, in this subsection, we propose a novel sparse precoder. On one hand, we can leverage optimization theory to adaptively allocate beams to UTs for enhanced transmission efficiency. On the other hand, by imposing a sparsity constraint on the total number of activated beams across

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all UTs, the overall complexity can be effectively reduced. Consequently, the sparse precoder achieves a flexible trade-off between performance and complexity, while encompassing the above beam structured precoder as a special case.

Define an ordered index set of candidate beams for UT k as $\hat{\mathcal{N}}_k$ with $\hat{N}_k = |\hat{\mathcal{N}}_k|$. The space domain precoder is restricted to have the structure $\mathbf{w}_k = \mathbf{V}\mathbf{S}_{\hat{\mathcal{N}}_k}\hat{\mathbf{w}}_k$, where $\hat{\mathbf{w}}_k \in \mathbb{C}^{\hat{N}_k}$ is the beam domain precoder. Then the ergodic rate upper bound in (21) becomes

$$\hat{\mathcal{I}}_k^{\mathrm{b,ub}} = \log\left(1 + \frac{\beta_k \tilde{\mathbf{g}}_k^{\mathrm{H}} \hat{\mathbf{Q}}_{k,k} \hat{\mathbf{w}}_k \hat{\mathbf{w}}_k^{\mathrm{H}} \hat{\mathbf{Q}}_{k,k}^{\mathrm{H}} \tilde{\mathbf{g}}_k}{\sum\limits_{i \in \mathcal{K}\setminus k} \beta_k \tilde{\mathbf{g}}_k^{\mathrm{H}} \hat{\mathbf{Q}}_{k,i} \hat{\mathbf{w}}_i \hat{\mathbf{w}}_i^{\mathrm{H}} \hat{\mathbf{Q}}_{k,i}^{\mathrm{H}} \tilde{\mathbf{g}}_k + \sigma_k^2}\right), \quad (25)$$

where $\hat{\mathbf{Q}}_{k,i} = \mathbf{S}_{\hat{\mathcal{N}}_k}^{\mathrm{H}} \mathbf{V}^{\mathrm{H}} \mathbf{V} \mathbf{S}_{\hat{\mathcal{N}}_i}$. We introduce the sparsity constraint on the beam domain precoder as

$$\sum_{k \in \mathcal{K}} \|\hat{\mathbf{w}}_k\|_0 \le N_{\mathrm{e}}, \quad (26)$$

which limits the sum of the number of non-zero elements in beam domain precoders of all UTs to a pre-set number N_{e}, and thus it also serves the function of beam allocation. Then the precoder design in terms of ergodic sum-rate upper bound maximization with sparsity constraint is given by

$$\max_{\{\hat{\mathbf{w}}_k\}_{k\in\mathcal{K}}} \sum_{k \in \mathcal{K}} \hat{\mathcal{I}}_k^{\mathrm{b,ub}} \quad (27)$$

$$\text{s.t.} \sum_{k \in \mathcal{K}} \hat{\mathbf{w}}_k^{\mathrm{H}} \hat{\mathbf{A}}_k \hat{\mathbf{w}}_k \le P, \quad (27a)$$

$$\sum_{k \in \mathcal{K}} \|\hat{\mathbf{w}}_k\|_0 \le N_{\mathrm{e}}, \quad (27b)$$

where $\hat{\mathbf{A}}_k = \mathbf{S}_{\hat{\mathcal{N}}_k}^{\mathrm{H}} \mathbf{V}^{\mathrm{H}} \mathbf{V} \mathbf{S}_{\hat{\mathcal{N}}_k}$.

Due to the nonconvex and complex sparsity constraint, directly solving the problem (27) is rather challenging. To address this multi-constrained optimization problem, we adopt penalty dual decomposition (PDD) method [12]. Specifically, by introducing auxiliary variables $\mathbf{a}_k = \hat{\mathbf{w}}_k, \forall k \in \mathcal{K}$, problem (27) is equivalently transformed into the following problem:

$$\max_{\{\hat{\mathbf{w}}_k, \mathbf{a}_k\}_{k\in\mathcal{K}}} \sum_{k \in \mathcal{K}} \hat{\mathcal{I}}_k^{\mathrm{b,ub}} \quad (28)$$

$$\text{s.t.} \sum_{k \in \mathcal{K}} \hat{\mathbf{w}}_k^{\mathrm{H}} \hat{\mathbf{A}}_k \hat{\mathbf{w}}_k \le P, \quad (28a)$$

$$\sum_{k \in \mathcal{K}} \|\hat{\mathbf{a}}_k\|_0 \le N_{\mathrm{e}}, \quad (28b)$$

$$\hat{\mathbf{w}}_k = \mathbf{a}_k, \quad \forall k \in \mathcal{K}. \quad (28c)$$

Then the augmented Lagrangian (AL) problem of (28) is defined as

$$\max_{\{\hat{\mathbf{w}}_k, \mathbf{a}_k\}_{k\in\mathcal{K}}} \sum_{k \in \mathcal{K}} \hat{\mathcal{I}}_k^{\mathrm{b,ub}} - \frac{1}{2\rho} \sum_{k \in \mathcal{K}} \|\hat{\mathbf{w}}_k - \mathbf{a}_k + \rho \mathbf{m}_k\|^2 \quad (29)$$

$$\text{s.t.} \sum_{k \in \mathcal{K}} \hat{\mathbf{w}}_k^{\mathrm{H}} \hat{\mathbf{A}}_k \hat{\mathbf{w}}_k \le P, \quad (29a)$$

$$\sum_{k \in \mathcal{K}} \|\hat{\mathbf{a}}_k\|_0 \le N_{\mathrm{e}}, \quad (29b)$$

where ρ is the penalty and $\mathbf{m}_k \in \mathbb{C}^{\hat{N}_k}$ is the dual variable for the constraint (28c) [12]. The PDD method solves problem (28) via a nested loop structure. Specifically, it updates the primal variables by solving the AL problem with block coordinate descent (BCD)-type methods in the inner loop, and adaptively updates the dual variables and the penalty parameters in the outer loop.

1) The subproblem w.r.t. $\hat{\mathbf{w}}_k$:

$$\max_{\{\hat{\mathbf{w}}_k\}_{k\in\mathcal{K}}} \sum_{k \in \mathcal{K}} \hat{\mathcal{I}}_k^{\mathrm{b,ub}} - \frac{1}{2\rho} \sum_{k \in \mathcal{K}} \|\hat{\mathbf{w}}_k - \mathbf{a}_k + \rho \mathbf{m}_k\|^2 \quad (30)$$

$$\text{s.t.} \sum_{k \in \mathcal{K}} \hat{\mathbf{w}}_k^{\mathrm{H}} \hat{\mathbf{A}}_k \hat{\mathbf{w}}_k \le P, \quad (30a)$$

This subproblem can be solved by a modified Algorithm 1, where $\tilde{\mathbf{w}}_k$ is replaced by

$$\hat{\mathbf{w}}_k = \left(\sum_{i \in \mathcal{K}} \tilde{q}_i |\gamma_i|^2 \beta_i \hat{\mathbf{Q}}_{i,k}^{\mathrm{H}} \tilde{\mathbf{g}}_i \tilde{\mathbf{g}}_i^{\mathrm{H}} \hat{\mathbf{Q}}_{i,k} + \mu \hat{\mathbf{A}}_k + \frac{1}{2\rho}\mathbf{I}\right)^{-1} \times \left(\tilde{q}_k \tilde{\gamma}_k \sqrt{\beta_k} \hat{\mathbf{Q}}_{k,k}^{\mathrm{H}} \tilde{\mathbf{g}}_k + \frac{1}{2\rho}\left(\mathbf{a}_k - \rho \mathbf{m}_k\right)\right). \quad (31)$$

2) The subproblem w.r.t. $\mathbf{a}_k$:

$$\min_{\{\mathbf{a}_k\}_{k\in\mathcal{K}}} \frac{1}{2\rho} \sum_{k \in \mathcal{K}} \|\hat{\mathbf{w}}_k - \mathbf{a}_k + \rho \mathbf{m}_k\|^2$$

$$\text{s.t.} \sum_{k \in \mathcal{K}} \|\hat{\mathbf{a}}_k\|_0 \le N_{\mathrm{e}}. \quad (32)$$

The optimal solution of problem (32) can be obtained as [13]

$$\mathbf{\Phi} = \mathcal{H}_{N_{\mathrm{e}}}\left(\hat{\mathbf{W}} + \rho \mathbf{M}\right), \quad (33)$$

where $[\mathbf{\Phi}]_{:,k} = \mathbf{a}_k$, $\left[\hat{\mathbf{W}}\right]_{:,k} = \hat{\mathbf{w}}_k$, $[\mathbf{M}]_{:,k} = \mathbf{m}_k$, and $\mathcal{H}_{N_{\mathrm{e}}}(\mathbf{X})$ denotes retaining the N_{e} largest elements of $\mathbf{X}$ and setting the rest to 0.

The proposed sparse precoder design based on PDD method is summarized in Algorithm 2.

Algorithm 2: Sparse Precoder Design

Input: $\beta_k, \tilde{\mathbf{g}}_k, \hat{\mathbf{A}}_k, \sigma_k^2, \hat{\mathbf{Q}}_{k,k'}, \forall k, k' \in \mathcal{K}, P, N_{\mathrm{e}}$

Initialize: $\hat{\mathbf{w}}_k, \mathbf{a}_k, \mathbf{m}_k, \forall k \in \mathcal{K}$, tolerance ε, penalty ρ, penalty update factor ξ.

repeat

  repeat

    Update $\hat{\mathbf{w}}_k, \forall k \in \mathcal{K}$ using modified Algorithm 1,

    Update $\mathbf{a}_k, \forall k \in \mathcal{K}$ using (33),

  until *convergence*;

  if $\max_{k\in\mathcal{K}} \|\hat{\mathbf{w}}_k - \mathbf{a}_k\|_\infty \le \varepsilon$ **then**

    $\mathbf{m}_k \leftarrow \mathbf{m}_k + \rho^{-1}\left(\hat{\mathbf{w}}_k - \mathbf{a}_k\right), \forall k \in \mathcal{K}$,

  else

    $\rho \leftarrow \xi\rho$,

  end

until *convergence*;

Output: Then the space domain precoder is obtained as $\mathbf{w}_k = \mathbf{V}\mathbf{S}_{\hat{\mathcal{N}}_k}\hat{\mathbf{w}}_k, \forall k \in \mathcal{K}$.

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C. Complexity Analysis

We first analyze the design complexity of the proposed precoders. The design complexity per iteration of beam structured precoder, which mainly comes from Step 5 in Algorithm 1, is $\mathcal{O}\left(\sum_{k\in\mathcal{K}}\left(N_k^2K+N_k^3\right)\right)$. Similarly, the design complexity per iteration of the sparse precoder, which mainly comes from Step 4 in Algorithm 2, is $\mathcal{O}\left(\sum_{k\in\mathcal{K}}\left(\hat{N}_k^2K+\hat{N}_k^3\right)\right)$.

After designing the precoder, we need to apply it to the data to generate the transmitted signal, which is referred to as the implementation of the precoder. Since both proposed precoders have the beam structure, the transmitted signal is given by

$$\sum_{k\in\mathcal{K}}\mathbf{w}_k x_k=\mathbf{V}\sum_{k\in\mathcal{K}}\mathbf{S}_{\hat{\mathcal{N}}_k}\hat{\mathbf{w}}_k x_k. \tag{34}$$

From (34), the implementation of the precoder can be divided into two parts: 1) perform the beam domain precoding for each UT, i.e., $\hat{\mathbf{w}}_k x_k$; 2) transform the sum of beam domain transmitted signals for all UTs into the space domain with the beam matrix $\mathbf{V}$, which can be efficiently calculated with fast Fourier transform (FFT), since $\mathbf{V}$ can be regarded as a partial DFT matrix. Thus the implementation complexity of beam structured precoders is $\mathcal{O}\left(N\log_2 N_{\mathrm{x}}+M_{\mathrm{x}}N_{\mathrm{y}}\log_2 N_{\mathrm{y}}+\sum_{k\in\mathcal{K}}N_k\right)$. For the sparse precoders, since only N_{e} elements are non-zero for the beam domain precoders of all UTs, the implementation complexity is $\mathcal{O}\left(N\log_2 N_{\mathrm{x}}+M_{\mathrm{x}}N_{\mathrm{y}}\log_2 N_{\mathrm{y}}+N_{\mathrm{e}}\right)$. As a comparison, the implementation complexity of the space domain precoder is $\mathcal{O}\left(MK\right)$.

Note that the space domain statistical CSI is constant at different subcarriers and varies slowly compared to the instantaneous CSI [4], [14]. Therefore, the designed precoders with the statistical CSI can be applied at different subcarriers for a long time, for example, 300 ms. Hence, the implementation complexity of the precoders dominates the total computational complexity, and the complexity of the proposed precoders can be much lower than that of space domain precoders due to the utilization of the beam domain channel sparsity in massive MIMO LEO satellite communications.

IV. Simulation Results

In this section, we provide simulation results to illustrate the performance of the proposed sparse precoder for massive MIMO LEO satellite communications. The main simulation parameters are listed in Table I. The UTs are generated in the coverage area of the satellite with uniform distribution. The satellite orbit radius is $R_{\mathrm{s}}=H+R_{\mathrm{e}}$, where H is the orbit altitude of the satellite and R_{e} is the earth radius. The elevation angle is calculated as $\alpha_k=\cos^{-1}\left(R_{\mathrm{s}}\sin\left(\vartheta_k\right)/R_{\mathrm{e}}\right)$ [15], where ϑ_k is the nadir angle of UT k. The distance between UT k and the satellite is $D_k=-\sin\left(\alpha_k\right)R_{\mathrm{e}}+\sqrt{2R_{\mathrm{e}}H+\left(\sin\left(\alpha_k\right)R_{\mathrm{e}}\right)^2+H^2}$ [16]. Let G_{ut} denote the antenna gain at UTs, and G_{sat} denote the per-antenna gain at satellites. For different elevation angles, the values of path loss and shadow fading are adopted based on the dense urban scenario in [16]. For the beam selection in the proposed precoders, we employ a diamond-shaped strategy. This choice is motivated by the observation that the energy distribution of the diffusion vector resembles the Kronecker product of two Dirichlet kernels, resulting in a diamond-shaped region of dominant energy concentration in the beam domain.

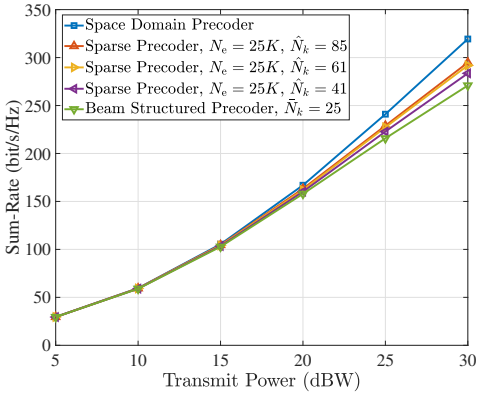


Fig. 1. Ergodic sum-rate performance of different precoders with different numbers of selected beams.

TABLE I
Simulation Parameters

Parameter	Value
Carrier frequency f_{c}	2 GHz
Number of satellite antennas $M_{\mathrm{x}}\times M_{\mathrm{y}}$	16×16
Number of UTs K	128
Bandwidth	10 MHz
Number of subcarriers	1024
Orbit altitude H	600 km
Earth Radius R_{e}	6378 km
Maximum nadir angle $\vartheta_{\max}$	30°
Antenna spacing	$\lambda/2$
Antenna gain G_{sat}, G_{ut}	5 dBi, 0 dBi

Fig. 1 shows the ergodic sum-rate performance of the proposed sparse precoder with different numbers of selected beams, $\hat{N}_k$. The space domain precoder in [5] is included for comparison. From the figure, as the set of candidate beams enlarges, the performance of the sparse precoder improves and gradually approaches that of space domain precoder. This is because a larger $\hat{N}_k$ leads to a higher degree of freedom for optimization, which ultimately improves the sum-rate performance. Moreover, although the sparse precoder utilizes the same total number of beams as the beam structured precoder and hence maintains identical implementation complexity, it achieves better performance by enabling efficient beam allocation among multiple UTs.

Fig. 2 shows the ergodic sum-rate performance of the proposed sparse precoder with different numbers of non-zero elements. From the figure, as the number of constrained non-zero elements in sparse precoders decreases, its performance deteriorates. However, the performance of sparse precoder with $N_{\mathrm{e}}=13K$ still outperforms the beam structured precoder with $\bar{N}_k=25$ while achieving approximately half the implementation complexity in the beam domain.

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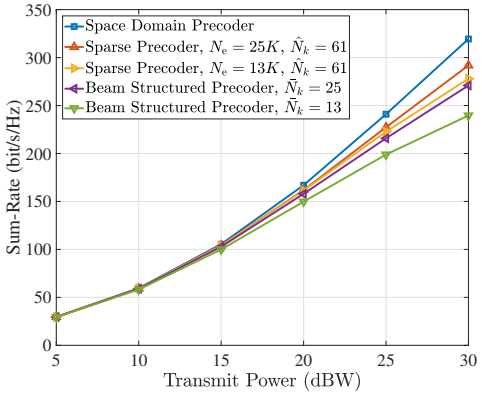


Fig. 2. Ergodic sum-rate performance of the different precoders with different numbers of non-zero elements.

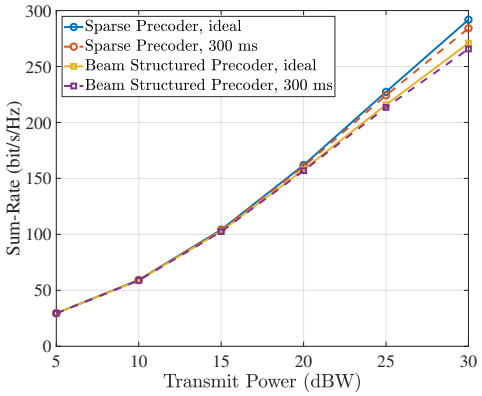


Fig. 3. Ergodic sum-rate performance of the different precoders for different update periods.

Fig. 3 shows the ergodic sum-rate performance of different precoders with different update periods. From the figure, the performance degrade of the proposed precoders is negligible with the update period of 300 ms.

Fig. 4 shows the complexities of different precoders, where the update period is 300 ms. The total complexity refers to the sum of the design complexity and the implementation complexity. From the figure, the proposed beam structured precoder and sparse precoder have much lower complexity than the space domain precoder due to the utilization of channel sparsity in the beam domain. Moreover, the sparse precoder with $N_{\mathrm{e}} = 13K$ can achieve lower complexity than the beam structured precoder while have a better performance as shown in Fig. 2.

V. Conclusions

In this paper, we have investigated DL precoding for massive MIMO LEO SATCOM. We developed a beam based channel model that captures the spatial single path property of LEO SATCOM channels and exposes their sparsity in the beam domain. Building on this channel model, we transformed the design of the space domain precoder into that of a lower-dimensional beam domain precoder. Furthermore, by imposing a sparsity constraint on the beam domain precoder, we proposed a sparse precoder that enables a flexible trade-off between system performance and computational complexity.

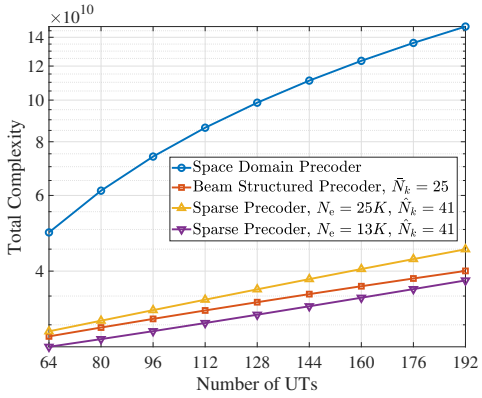


Fig. 4. Complexities of different precoders versus the number of UTs.

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